

FRANKLINE MISANGO OYOLO

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SUMMARY

Quantitative Analyst and Founder with expertise in multi-asset trading infrastructure, machine learning, and client-centric financial innovation across equities, futures, and options.

EDUCATION

The University of Hong Kong

Bachelor of Engineering in Artificial Intelligence— GPA: 3.73/4.3

Expected Dec 2027

Hong Kong SAR

CURRENT ENGAGEMENT

Arithmax Research

Principal Founder

Jan 2023 – Present

Remote

- Leads quantitative research and backtesting across momentum, macro, and market microstructure strategies, publishing investment memos and technical papers.
- Designed low-latency C++ and Rust trading infrastructure for robust signal-to-execution workflows and sub-millisecond event handling.
- Integrates alternative data and FPGA simulation workflows to improve regime detection, model resilience, and production-readiness.

PROFESSIONAL EXPERIENCE

Completed engagements

Citi

Markets Summer Analyst (Sales and Trading)

Jun 2025 – Aug 2025

Hong Kong SAR

- Selected for a dual-rotation program across trading desks, delivering cross-asset analysis under fast market conditions.
- Partnered with global teams to operationalize factor-model insights and present desk-level recommendations to senior management.
- Automated desk workflows with VBA, Python, and Excel, improving execution speed and reducing manual error during high-volume sessions.

The Swiss Federal Institute of Technology (EPFL)

Doctoral Research Assistant

Jul 2025 – Aug 2025

Lausanne, Switzerland

- Awarded the Summer @ EPFL Grant (CHF 5,000) in a globally selective 0.8% acceptance cohort.
- Built a multi-scale Python feature pipeline combining demographic and transaction signals within a 1,000m station radius.
- Improved RMSE by 27% over ST-GCN baselines (plus 13.4% from OSM integration), enabling more reliable demand forecasts for fiscal planning and infrastructure investment decisions.

BASF SE

Machine Learning Engineer

Jan 2025 – Jun 2025

Hong Kong SAR

- Built Python DiD and local-projection shock models with 50k Monte Carlo paths, reducing DSO volatility by 42%.
- Optimized supplier networks via CVaR-constrained simulation, reducing landed-cost variance to 25 bps and unlocking \$2.3M in working capital.

Manulife Financial Corporation

Applied Data Scientist

Jun 2024 – Aug 2024

Hong Kong SAR

- Built a Python metadata enrichment engine that raised actuarial tag coverage across 2.3M Azure Purview assets from 4% to 65%.
- Automated 300+ data quality checks and lineage workflows, improving new business margin screening precision to 92% for IFRS 17 reporting.

MARKETS APTITUDE

Cross-Regime Performance Analysis of an Algorithmic Strategy for a Diversified Leverage Index Fund Portfolio Caltech Investment Management

- Developed a risk-parity portfolio model using stochastic control equations to achieve 46.2 percent annualized returns.
- Mitigated volatility decay in leveraged ETFs via GARCH optimized weighting, reducing decay by 38 percent versus naive allocation.

Exploiting Arbitrage in Currency Crashes: The R-Zone Early Warning Strategy

HKU Trading Group

- Developed a logistic regression classifier with an 84 percent accuracy rate to identify FX regime shifts and predict currency crashes.
- Utilized FX-implied volatility surfaces to engineer hedging strategies, mitigating drawdown by 24 percent during market dislocations.

Intraday Momentum Breakout Strategy: A Volatility-Targeted Approach to E-mini Futures Trading HKU Quantitative Society

- Built an intraday momentum breakout strategy with volatility targeting and adaptive position sizing for E-mini futures.
- Improved risk-adjusted performance through regime-aware entry and stop logic, reducing intraday drawdown and improving execution consistency.

PROGRAMMING PROJECTS

FeenQR: Automated Portfolio Analysis Application | ★ 142 28

C#, Python, AI

- Engineered an end-to-end automation platform for portfolio stress-testing and rebalancing across 40 distinct asset classes.
- Integrated real-time regime detection and neural network volatility forecasting to automate risk management for institutional-grade strategies.

TradeKernel: CUDA-Accelerated Market Making | ★ 310 56

C++, CUDA, High-Performance Computing

- Implemented GPU-accelerated finite-difference solvers for a 136x speedup, preserving 97.4 percent PnL efficiency during execution.
- Built an automated order flow toxicity tracker to reduce adverse selection by 41 percent through real-time imbalance-aware quoting.

SKILLS, INTERESTS & ACTIVITIES

Research: Options Pricing (Black-Scholes, Heston), Stochastic Calculus (HJB), Order Book Dynamics (LOB), PDE Solvers (Crank-Nicolson, FD)

Languages: English (Native), German (A2), French (A2), Mandarin (Beginner)

Programming: Python, C#, C++, Rust, CUDA (GPU Acceleration), VHDL/Verilog (FPGA)

Tools: MS Excel, VBA (Macro Automation), Bloomberg Terminal, Reuters Eikon

Certifications: AWS Machine Learning Specialty, AWS Solutions Architect Professional, CFI Financial Modeling and Valuation Analyst (FMVA)

Extra Curricular: Chess (Blitz 1800, Rapid 1500), Leetcode (top 5%), Cycling, International travel

Self Readings : Financial Markets, Global Geopolitics, Chip Wars, Planetary Science, Evolutionary Biology, Zen Buddhism

Campus Involvement: HKU Trading Group TMT Software Developer, HKU Quantitative Researcher for Analysis of Algorithms

Awards: Amazon Web Services ABW Grant (2024, 2025), Python Software Foundation Grant (2024)